DELUXE STORE BUSINESS ANALYSIS REPORT

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1. Introduction and Business Objectives

1.1 Introduction

This project analyses the Deluxe Store dataset covering transactions from 2014 to 2017 to evaluate the store's sales performance, profitability, product performance, customer segments, regional performance, and discount patterns.

Microsoft Excel was used to clean and prepare the dataset, while Microsoft Power BI was used to analyze the data and develop an interactive dashboard. The analysis aims to identify business strengths, areas of concern, and opportunities for improving profitability and decision-making.

1.2 Business Objectives

The main objectives of this project are to:

- Evaluate sales and overall business performance between 2014 and 2017.
- Identify product categories, sub-categories, and individual products that contribute to or reduce profitability.
- Compare profitability across customer segments and geographical regions.
- Examine how discount levels relate to profitability and identify opportunities to improve discount management.

1.3 Business Questions

The analysis addresses four key business questions:

1. How did sales and overall business performance change between 2014 and 2017?
2. Which product categories, sub-categories, and individual products contributed most to profit, and which generated losses?
3. Which customer segments and geographical regions contributed most to sales and profit?
4. How were discount levels associated with profitability, and what actions could help improve business performance?

2 Analytical Approach

The analysis used the Deluxe Store dataset covering transactions from 2014 to 2017. The dataset was cleaned and validated in Microsoft Excel and Power Query, resulting in 9,993

rows and 24 columns. Microsoft Power BI was then used to analyze sales, profitability, products, customer segments, regions, and discount patterns through measures and interactive visualizations.

Detailed data-cleaning procedures and decisions are documented separately in the Data Preparation Documentation.

3. Key Findings and Business Insights

3.1 Overall Sales and Business Performance

The analysis of the Deluxe Store dataset for 2014–2017 showed total sales of approximately $2.30 million and total profit of $286,408.08, resulting in a profit margin of 12.47%.

Monthly sales fluctuated throughout the period. September 2014, November 2015, December 2016, and November 2017 recorded the highest sales months in their respective years, while January and February generally showed weaker performance.

Technology generated the highest overall profit among the product categories, while Furniture was the weakest-performing category. In 2017, Furniture recorded zero profit, highlighting the need to investigate its sales, costs, pricing, and discount practices.

3.2 Product Profitability

The analysis revealed differences in profitability across product sub-categories and individual products. Phones generated the highest profit among the selected top-performing sub-categories in 2014 and 2015, while Copiers led in 2016 and 2017. This indicates that Copiers were a strong source of profit in the later years of the analysis period.

However, some sub-categories recorded losses. Tables generated the largest loss, at approximately $17,725, followed by Bookcases at approximately $3,473 and Supplies at approximately $1,189.

At the individual product level, the Canon imageCLASS 2200 Advanced Copier was the highest-profit product, generating $25,200. In contrast, the Cubify CubeX 3D Printer Double Head Print recorded the largest loss, at approximately $8,880. The Bush Advantage Collection Racetrack Conference Table also recorded a loss of approximately $1,934.

These findings highlight the importance of identifying consistently profitable products and investigating loss-making products through a review of their costs, selling prices, and discount levels.

3.3 Customer Segment and Regional Performance

The analysis showed that the Consumer segment generated the highest profit, at approximately $134,119, while Home Office generated the lowest profit, at approximately $60,311. This suggests that Consumer customers were an important contributor to the store's profitability during the period analysed.

Regional analysis showed that the West region generated the highest profit, at approximately $108,418, while the Central region recorded the lowest profit, at approximately $39,706. The differences between regions suggest opportunities to investigate regional sales patterns, customer needs, product preferences, and operating costs.

The business could use these findings to understand the factors behind the strong performance of the Consumer segment and West region, while investigating opportunities to improve profitability among Home Office customers and in the Central region.

3.4 Discount Patterns and Profitability

The analysis showed that transactions with no discount (0%) generated the highest total profit, approximately $320,988. Several discounted groups recorded considerably lower profits. Binders had the highest average discount at 37.23%, while Labels had the lowest at 6.87%.

These findings suggest that discount practices should be reviewed to determine whether discounts are contributing to reduced profitability. However, further analysis of sales volume, product costs, selling prices, and profit margins is needed before concluding that higher discounts directly cause lower profits. The business should assess whether discounts generate enough additional sales to justify their impact on profitability.

4. Recommendations

Based on the findings of the analysis, the following actions are recommended:

1. Improve Sales During Weaker Periods: Investigate the reasons for lower sales in January and February and consider appropriate strategies to improve performance during these months.

2. Review Loss-Making Sub-Categories: Examine Tables, Bookcases, and Supplies to identify the causes of their losses. Review their costs, selling prices, and discount practices to identify opportunities to improve profitability.

3. Strengthen Product Profitability: Maintain focus on profitable products, particularly the Canon image CLASS 2200 Advanced Copier, and investigate opportunities to improve the profitability of other products.

4. Improve Customer Segment and Regional Performance: Study purchasing patterns in the Home Office segment and investigate the factors contributing to the Central region's lower profit. Use these insights to identify suitable improvement strategies.

5. Evaluate Discount Practices: Review the effectiveness of discounts, particularly for Binders, which had the highest average discount of 37.23%. Assess whether discounts generate sufficient additional sales to justify their impact on profitability.

6. Monitor Business Performance: Use the Power BI dashboard to regularly monitor sales, profit, profit margin, customer segments, regional performance, and discount patterns to support informed business decisions.

5. Limitations

This analysis was based on historical transaction data from 2014 to 2017; therefore, the findings may not reflect current business performance.

The analysis identified patterns and relationships but did not establish direct causes for differences in sales and profitability.

Further investigation into product costs, pricing, customer purchasing behavior, and discount practices would provide deeper insights.

Additionally, the dataset did not contain a confirmed returns indicator, so actual product return rates could not be assessed.

6. Conclusion

The analysis of the Deluxe Store dataset for 2014–2017 revealed total sales of approximately $2.30 million and total profit of $286,408.08, with a profit margin of 12.47%. Technology was the strongest-performing product category, while Furniture showed weaker profitability.
The analysis also identified loss-making products and sub-categories, differences in customer segment and regional performance, and patterns in discount-related profitability.

These findings highlight opportunities to improve profitability by reviewing loss-making products, evaluating discount practices, investigating weaker-performing segments and regions, and monitoring key performance indicators. The Power BI dashboard provides a visual tool to support informed business decisions and ongoing performance monitoring